

PC CONTROL WITH EYE MOVEMENT For Physically Challenged

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Millions of people across the globe suffer from neuro conjugate disorder, or some other form of paralysis, which does not let them move any body part except their head and the eyes. To overcome this challenge, we can make a smart personal computer (PC) that empowers such physically challenged persons to control a computational system with the help of

facial movements. It enables control using face and eye movements and works even in low light or dark conditions.

Coordinate movements in the PC = (1080,720) to (0,0)

where $x = 1080$, $y = 720$

$mapx = (x)$, $mapy = (y)$

The solution is based on three different functions:

1. Blinking detection of the right

eye for system operation

2. Detection of the eye movement using image processing

3. Translation of the eye movement and blink to control graphic user interface (GUI) of the PC

The challenges

The above three functions come with several challenges, which can be handled with various steps for the device to work perfectly. These are:

Differentiating between natural and intentional eye blink. Our eyes involuntarily blink at regular intervals so that they remain moist and free of dust. To perform a left or right mouse click, here an eye will have to blink intentionally. However, there has to be some way to distinguish between involuntary and intentional eye blinks so that the device functions properly.

To do that, two sensors (one corresponding to each eye) can be integrated in such a way that when both of them detect an eye blink at the same time, it is considered a natural eye blink. If the blinking happens in only one eye (either left or right), it is considered as an intentional eye blink, resulting in a left or right

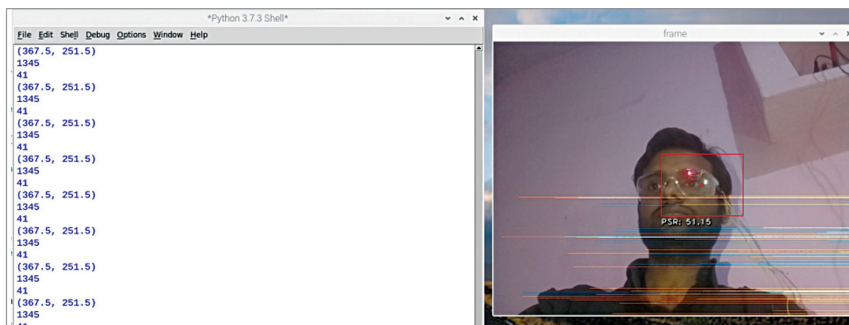


Fig. 1: Author's eye movement data converted to coordinates

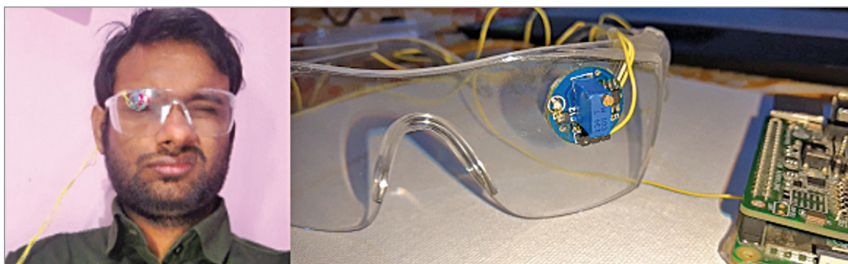


Fig. 2: Author testing the blink sensor

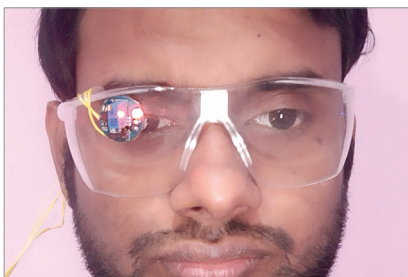


Fig. 3: The author with his prototype eyeglasses

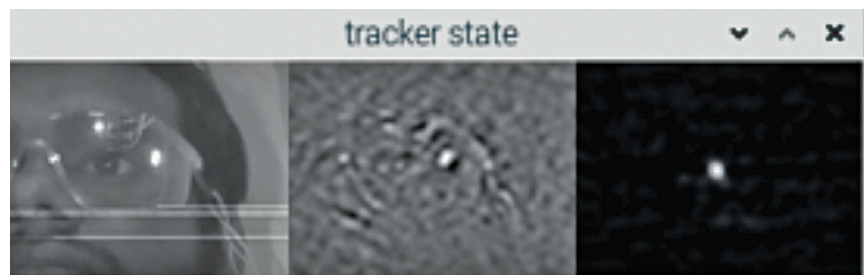


Fig. 4: Author's eye image processing